ONE-PAGE BRIEF

Post-quantum cryptography review: arc-node @ 6e764023ee65

Client: Circle (Arc) — public repository (passive review) · Source: github.com/circlefin/arc-node, commit `6e764023ee65` · Reviewed: 2026-11-06 15:31 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY

Level 2 of 4

post-quantum in use outside the critical paths

FINDINGS

0 Critical **17** High **20** Medium **31** Low **61** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)

Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Extend post-quantum protection to the critical roles that stay classical, starting with what can be recorded today (wrapped keys, channel key exchange).

WHAT MATTERS MOST

High QSI-2026-19526	**rmcp 1.4.0 is affected by GHSA-33f5-2c5q-wgwj (CVE-2026-63127)** What to do: Upgrade to a fixed version (2.0.0).
High QSI-2026-19528	**rmcp 1.4.0 is affected by GHSA-9pj6-vhgr-3mwh (CVE-2026-63128)** What to do: Upgrade to a fixed version (2.0.0).
High QSI-2026-19441	**Possible hard-coded secret in...** preliminary What to do: Move secrets to a secrets manager/environment; rotate the exposed values.
High QSI-2026-19499	**hickory-proto 0.25.2 is affected by GHSA-3v94-mw7p-v465 (RUSTSEC-2026-0118)** What to do: Upgrade to a fixed version (0.26.0-beta.1).

64 more in the full report.

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium transaction and message signing	ECDSA, Ed25519	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum...
Medium role not determined	Ed25519, ECDSA	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)...

KNOWN VULNERABLE COMPONENTS

Advisories for the versions in use: 45; unlikely to apply to this build: 4 (a Cargo feature that is off, affected functions not named in the code, or a platform that is not built); to review: 41.

Static review of source code and manifests; files analysed: 1135. No requests were sent to the owner's systems.

Full report, content SHA-256: `2cb1f259975c8af684ffa81af50b3a9c029ff2d5a6375299ede323d6502412cd`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.